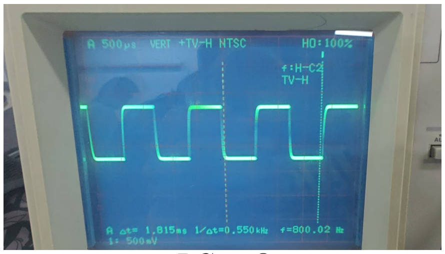
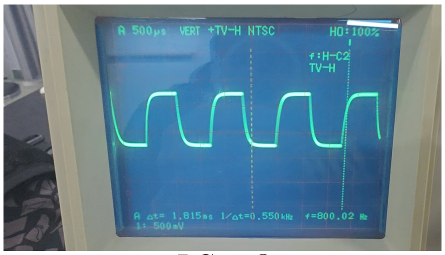
0.1 电流图像

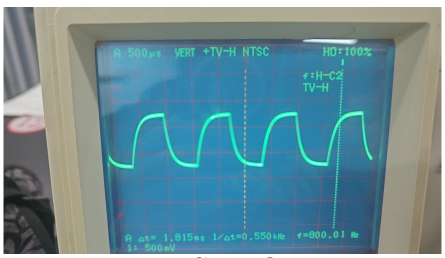
0.1.1 RC



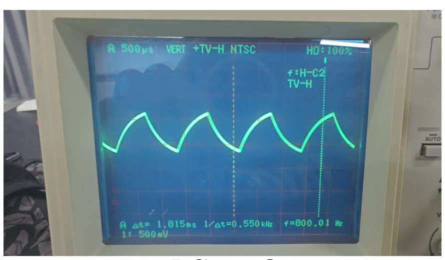
$RC100\Omega$



$RC510\Omega$

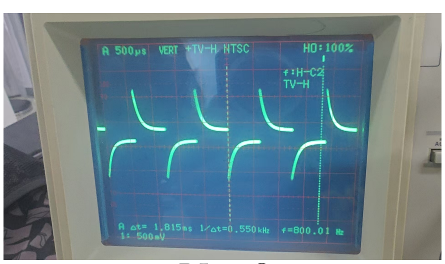


$RC1000\Omega$

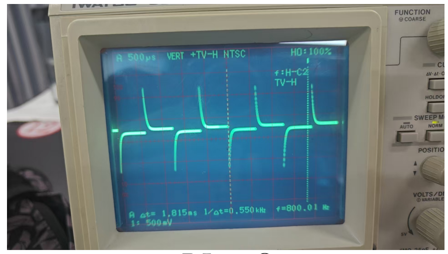


$RC3000\Omega$

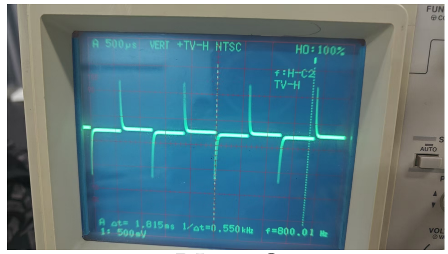
0.1.2 RL



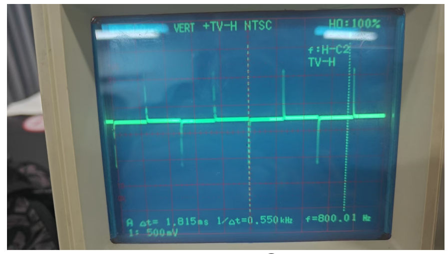
$RL100\Omega$



$RL510\Omega$

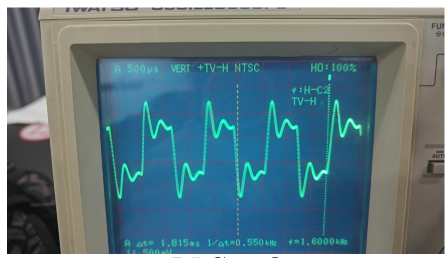


$RL1000\Omega$

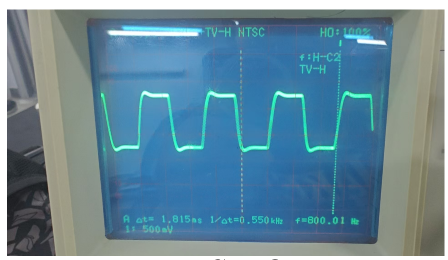


$RL3000\Omega$

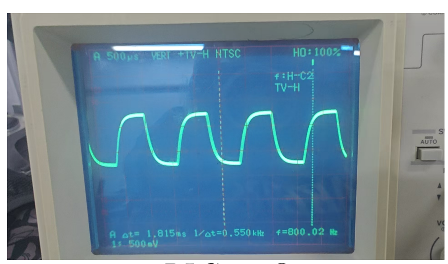
0.1.3 RLC



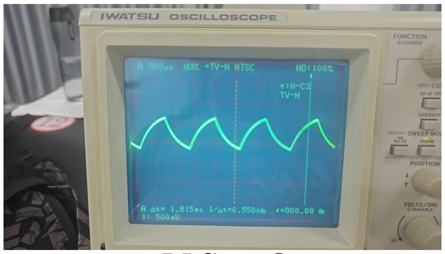
$RLC100\Omega$



$RLC510\Omega$



$RLC1000\Omega$



$RLC3000\Omega$

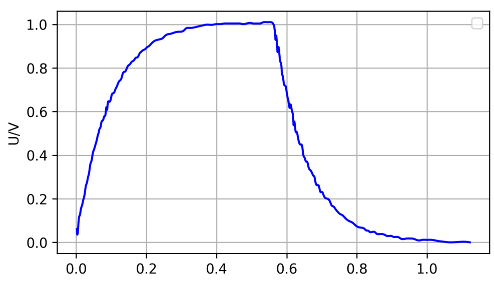
我们实验的参数为 $C = 0.1\mu F, L = 20mH$.

$$4\frac{L}{C} = \frac{4 \times 20 \times 10^{-3}H}{1 \times 10^{-7}F} = 8 \times 10^5 \Omega^2$$

当电阻 $R = 510\Omega$和$R = 100\Omega$ 时，电路大致处于欠阻尼状态；当 $R = 1000\Omega$和$R = 3000\Omega$ 时，电路处于过阻尼状态.

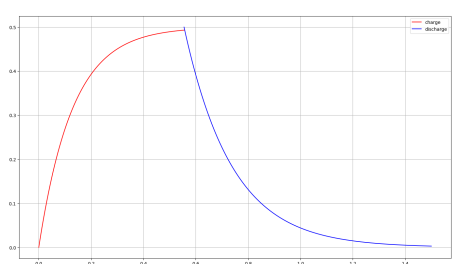
以 894.427Ω 作为标准，将波形划分为欠阻尼，临界阻尼和过阻尼状态.

0.2 数据拟合



数据拟合

在充电过程中，我们可以读出 $T_{0.5} = 0.09ms(U_C = 0.632U)$，在放电过程中，我们可以读出 $T_{0.5} = 0.0955ms(U_C = 0.368U)$，然后我们利用这两个数据重新画图.



重新绘制

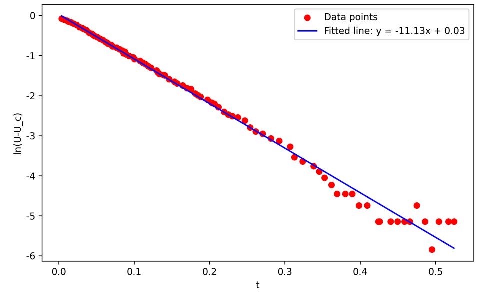
理论计算：

$$T_{0.5} = \tau ln2 = RCln2 = 1000\Omega \times 1 \times 10^{-7}F = 0.1ms$$

0.3 指数拟合

我们也可以通过指数拟合得到结果.

0.3.1 充电过程



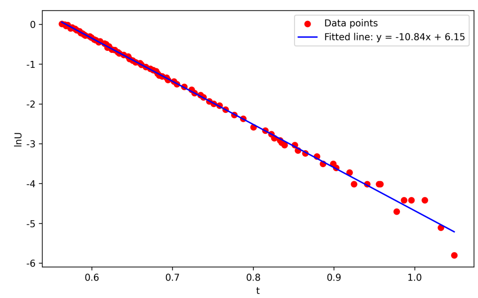
充电过程

得到

$$T_{0.5} = \frac{1}{11.13}ms = 0.089847ms$$

$$\text{相对误差：} \frac{0.1 - 0.089847}{0.1} \times 100\% = 10.153\%$$

0.3.2 放电过程



放电过程

得到

$$T_{0.5} = \frac{1}{10.84}ms = 0.0922509ms$$

$$\text{相对误差：} \frac{0.1 - 0.0922509}{0.1} \times 100\% = 7.7491\%$$